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CT for Evaluation of Hemoptysis

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Abbreviation: DAH = diffuse alveolar hemorrhage

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the appropriate CT protocol for evaluation of patients with hemoptysis.
- Identify normal and abnormal bronchial artery anatomy.
- Discuss common vascular and nonvascular causes of hemoptysis.

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Hemoptysis, which is defined as expectoration of blood from the alveoli or airways of the lower respiratory tract, is an alarming clinical symptom with an extensive differential diagnosis. CT has emerged as an important noninvasive tool in the evaluation of patients with hemoptysis, and the authors present a systematic but flexible approach to CT interpretation. The first step in this approach involves identifying findings of parenchymal and airway hemorrhage. The second step is aimed at determining the mechanism of hemoptysis and whether a specific vascular supply can be implicated. Hemoptysis can have primary vascular and secondary vascular causes. Primary vascular mechanisms include chronic systemic vascular hypertrophy, focally damaged vessels, a dysplastic lung parenchyma with systemic arterial supply, arteriovenous malformations and fistulas, and bleeding at the capillary level. Evaluating vascular mechanisms of hemoptysis at CT also entails determining if a specific vascular source can be implicated. Although the bronchial arteries are responsible for most cases of hemoptysis, nonbronchial systemic arteries and the pulmonary arteries are important potential sources of hemoptysis that must be recognized. Secondary vascular mechanisms of hemoptysis include processes that directly destroy the lung parenchyma and processes that directly invade the airway. Understanding and employing this approach allow the diagnostic radiologist to interpret CT examinations accurately in patients with hemoptysis and provide information that is best suited to directing subsequent treatment.

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Hemoptysis is defined as expectoration of blood from the alveoli or airways of the lower respiratory tract. Clinically, hemoptysis must be distinguished from hematemesis and aspirated blood related to nasopharyngeal injury. Although hemoptysis is commonly categorized in the literature as massive or submassive on the basis of the volume of expectorated blood, these terms lack a consensus definition, are often inaccurate, and have limited implications in management. Asphyxiation rather than exsanguination is the main cause of death in patients with hemoptysis. As little as 250 mL of blood, or less than the volume of a 12-oz soft drink, fills the entire tracheobronchial tree (1,2). In cases of life-threatening hemoptysis, the initial priority is given to maintaining the airway, optimizing oxygenation, and ensuring hemodynamic stability. After these initial goals are met, attention is then turned to determining the underlying cause and best available treatment options.

Causes of hemoptysis vary depending on the geographic location and the practice setting to which a patient presents. Although tuberculosis remains the most common cause of hemoptysis in developing countries, other respiratory infections, malignant lesions, chronic bronchitis, and bronchiectasis are the most common causes of hemoptysis seen in tertiary referral centers in North America and Europe. Given

TEACHING POINTS

- A CT scanning protocol for patients who present with symptoms of hemoptysis should be used during the systemic arterial phase, with a contrast material injection rate of 4–5 mL/sec, with bolus tracking triggered at the middle descending thoracic aorta.
- No single pathophysiologic mechanism causes hemoptysis. Broadly, the mechanisms can be divided between primary vascular and secondary vascular causes. Primary vascular mechanisms include chronic systemic vascular hypertrophy, focally damaged vessels, dysplastic lung parenchyma with systemic arterial supply, arteriovenous malformations and fistulas, and capillary-level bleeding. Mechanisms in which vessels are secondarily involved include those that directly destroy the parenchyma such as infections and infarcts and those that involve direct invasion of the airway by a tumor or benign process such as a broncholith. Some underlying disease processes can produce hemoptysis by means of multiple mechanisms.
- When bronchial arteries are hypertrophied, cardiac output to them can increase from a normal output of 1% to up to 30%. Furthermore, these hypertrophied arteries are thin walled and fragile and are exposed to systemic pressure, making them prone to rupture into the airways, which can cause hemoptysis.
- Although bronchial arteries may contribute to increased systemic vascular supply via hypertrophied normal anatomic pathways, nonbronchial systemic arteries can contribute increased systemic vascular supply via acquired anatomic pathways.
- Pseudoaneurysms are more likely to originate in the pulmonary arterial tree, although bronchial artery and nonbronchial systemic pseudoaneurysms can occur.

that potential causes of hemoptysis are variable in both pathophysiologic mechanisms and imaging manifestations, hemoptysis remains a challenging topic for clinicians and radiologists. Further complicating the topic, many of the potential causes result in hemoptysis in only a small minority of cases.

CT has emerged as an important noninvasive tool in the evaluation of patients with hemoptysis and is rated as "usually appropriate" for both life-threatening and non–life-threatening hemoptysis in the American College of Radiology (ACR) appropriateness criteria (3). Many patients presenting with hemoptysis have known underlying conditions such as cystic fibrosis or tumors that can cause hemoptysis; in these cases, the role of CT is to evaluate for evidence of a new localizable or treatable source to best guide therapy. However, in other patients, hemoptysis may be an initial presenting symptom, without a known diagnosis. In these cases, CT can be used to evaluate for hemorrhage and a source but can also help to determine the cause or differential diagnosis. Once a suspected underlying cause of hemoptysis is identified, management options include combinations of medical, bronchoscopic, surgical, and endovascular therapies.

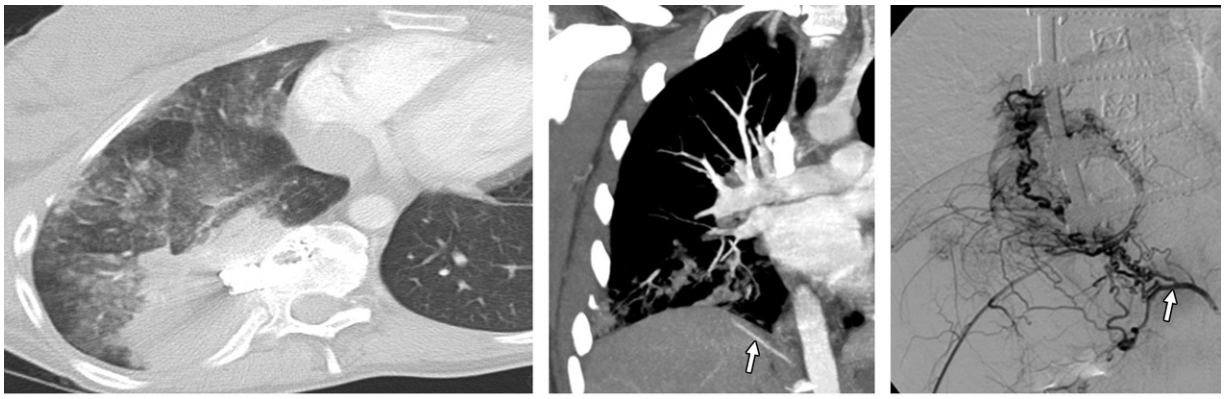
In this article, we discuss a systematic yet flexible approach to evaluation of patients with hemoptysis that includes consideration of the clinical scenario and relevant clinical history, focusing on the CT findings and pathophysiologic mechanisms. The first step in this approach involves identifying findings of hemorrhage at CT. The second step involves determining the mechanism of hemoptysis and whether a specific vascular supply can be implicated. This mechanism can then be correlated to an underlying diagnosis or differential diagnosis, when appropriate.

CT Scanning Technique

A CT scanning protocol for patients who present with symptoms of hemoptysis should be used during the systemic arterial phase, with a contrast material injection rate of 4–5 mL/sec, with bolus tracking triggered at the middle descending thoracic aorta (4). Acquisition during the systemic arterial phase ensures opacification of the bronchial arteries, which are responsible for bleeding in up to 90% of patients with hemoptysis who require endovascular intervention (1,3,5–8). The pulmonary arteries, which are responsible for bleeding in a minority of cases, are also reliably opacified in this phase, provided that a saline chaser is not used. A saline chaser should be avoided in patients with hemoptysis, because it may dilute or wash out the contrast material in the pulmonary arterial tree. In addition, ideally, a larger volume of iodinated contrast material (110–125 mL) is used to allow persistent pulmonary artery opacification at the time of the aortic trigger. Imaging coverage should extend from the lower neck through the celiac axis. Celiac axis coverage is recommended because some sources of hemoptysis may arise from upper abdominal vessels, including the inferior phrenic artery and the other celiac axis branches (9) (Fig 1).

For review of vascular and soft-tissue findings, images are reconstructed with a section thickness of 1 mm by using a standard kernel, and coronal and sagittal multiplanar reconstructions are made directly on the scanner from the raw data. Postprocessing software that is capable of generating double oblique and curved reconstructions, maximum intensity projection images, and three-dimensional volumetric reconstructions can be used to optimally show the vascular supply to areas of hemorrhage. For evaluation of the lung parenchyma, dedicated lung windows with a tailored kernel should also be used routinely.

For most patients who present with hemoptysis, imaging during a single systemic arterial phase should suffice. A noncontrast scan may be added in patients who have undergone surgery or who have experienced penetrating trauma



a. b. c.

Figure 1. Hemoptysis in a 24-year-old woman with a history of scoliosis who underwent spinal fusion surgery. **(a)** Axial CT image of the chest (lung window) shows signs of parenchymal hemorrhage, including extensive ground-glass opacity in the right middle and lower lobes. The ground-glass opacity surrounds the opacified medial right lower lobe, which is adjacent to spinal fusion instrumentation that protrudes directly into the lung. **(b)** Coronal multiplanar reformation image shows that this consolidated lung is served by enlarged systemic arteries arising from the inferior phrenic artery (arrow). These findings are consistent with systemic arterial supply to an intralobar pulmonary sequestration, which the patient likely acquired because of chronic inflammation related to the protruding spinal fusion instrumentation. **(c)** Catheter angiogram through the celiac axis shows the inferior phrenic artery (arrow) and allows confirmation of these findings. This patient was treated with polyvinyl alcohol particle embolization before undergoing surgical resection, at which the diagnosis of intralobar sequestration was confirmed. This case highlights the need to image through the celiac axis, because feeding arteries arising below the diaphragm can be an important cause of hemoptysis.

to allow differentiation of dense foreign bodies or surgical material from contrast-enhancing vessels or extravasated contrast material. In patients who have contraindications to the use of iodinated contrast agents, including a history of anaphylaxis, a risk-benefit assessment of the use of contrast material should be performed. In time-dependent situations involving patients with life-threatening hemoptysis, it may be advisable to administer contrast material under the observation of a resuscitation team or with the patient already intubated. Another approach involves performing a noncontrast scan to evaluate for localizable parenchymal findings of hemorrhage or an obvious cause of the hemoptysis. After review of the noncontrast images, a decision can be made regarding the necessity of a subsequent scan with intravenous contrast material to better evaluate for a diagnosis or plan for intervention.

As with adding a noncontrast acquisition, further modifications to CT scanning protocols for patients presenting with hemoptysis can be made on the basis of patient-specific factors and can be employed on a case-by-case basis. For example, adding a portal venous phase acquisition may be of value in patients with known tumors or infections, because these abnormalities may not adequately enhance in the systemic arterial phase. The portal venous phase scan can be performed with a fixed delay at 70 seconds. In patients with known conditions that affect the pulmonary arteries such as hereditary hemorrhagic telangiectasia, a prolonged bolus may be considered to ensure optimal pulmonary arterial opacification.

A summary of CT techniques for evaluation of hemoptysis is presented in Table 1.

CT Findings of Hemorrhage

The first step in CT interpretation for patients with hemoptysis is identifying findings suggestive of blood in the alveoli or airways. Blood in the alveoli typically manifests as ground-glass opacity or consolidation. Although these findings of airspace disease can be seen in other processes such as edema or pneumonia, in a patient presenting with hemoptysis, these findings should be regarded as potential areas of parenchymal hemorrhage. When hemorrhage is subacute, septal lines may also be present as macrophages enter the area and migrate to lymphatic vessels in the interlobular septa to remove the blood products (4,10,11) (Fig 2).

Patients who present with hemoptysis may also have findings of blood products in large or small airways. Blood products in the bronchial tree may manifest as detectable filling defects at CT. When they are in the larger airways, these filling defects show higher attenuation (>40 HU) in some cases, which allows them to be distinguished from mucus. Obstruction of larger airways with blood products may result in atelectasis. Blood products in the more distal airways typically manifest as centrilobular nodules that may have a tree-in-bud appearance. Findings of blood products in the airways can also be used to localize hemorrhage. Because airway blood products are often the result of distal aspiration, careful scrutiny for lesions or vascular sources that are associated with the more proximal bronchial tree is necessary (Fig 3).

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Table 1: CT Protocol for Evaluation of Hemoptysis

Imaging Parameter	Details
Contrast material (mL)	110–125
Contrast material injection rate (mL/sec)	4–5
No. of acquisitions	1
Timing of bolus tracking	Middle descending aorta
Coverage	Lower neck through the celiac axis
Reconstructions	
Vascular and soft tissue	
Planes of imaging	Axial, coronal, sagittal
Section thickness (mm)	1
Kernel	Standard
Lungs	
Section/interval thickness (mm)	3/2
Kernel	Dedicated lung
Postprocessing	Performed as needed with dedicated software
Double-oblique reconstruction	. . .
Curved reconstruction	. . .
Maximum intensity projection	. . .
Three-dimensional volumetric reconstruction	. . .
Optional sequences	
Noncontrast sequence	Consider in patients with penetrating trauma, prior surgery, or a history of anaphylaxis after iodinated contrast–enhanced sequence
Portal venous phase, with fixed 70-second delay	Consider in patients with tumors or pneumonia, because parenchymal and tumor enhancement are optimized in this phase

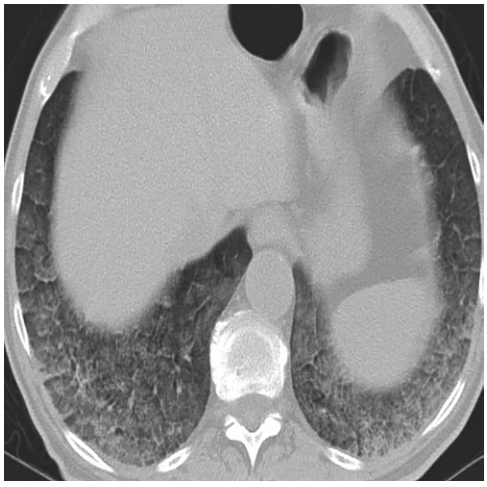


Figure 2. Small-vessel vasculitis and intermittent hemoptysis in a 38-year-old man. Axial CT image shows ground-glass opacity and interlobular septal lines in a "crazy-paving" pattern. The ground-glass opacity is an indirect sign that is consistent with pulmonary hemorrhage in patients with hemoptysis. The diagnosis was confirmed at bronchoscopy. Interlobular septal line thickening may be present in subacute pulmonary hemorrhage because macrophages enter the area to remove the blood products.

Blood products in the lungs or airways can also be used to direct targeted therapies; therefore, the distribution of hemorrhage is important. A diffuse pattern of pulmonary parenchymal hemorrhage is suggestive of a systemic process, and patients with this pattern are typically poor candidates for targeted interventions. In contradistinction, focal findings of parenchymal hemorrhage can be used to localize bleeding to a specific lesion or vascular territory. Although it is often difficult, distinguishing parenchymal from aspirated blood may be of value because aspirated blood suggests a more proximal or localized source. Both parenchymal and aspirated blood can be nodular, but findings of smaller well-defined clustered tree-in-bud centrilobular nodules favor aspiration from a proximal source, whereas regional or diffuse centrilobular ground-glass opacities are more suggestive of parenchymal bleeding. Temporal changes are also helpful. In patients who undergo serial examinations, particularly those with underlying chronic diseases or malignancies, new areas of airspace disease with hemoptysis can also be used to implicate areas of active or recent hemorrhage.

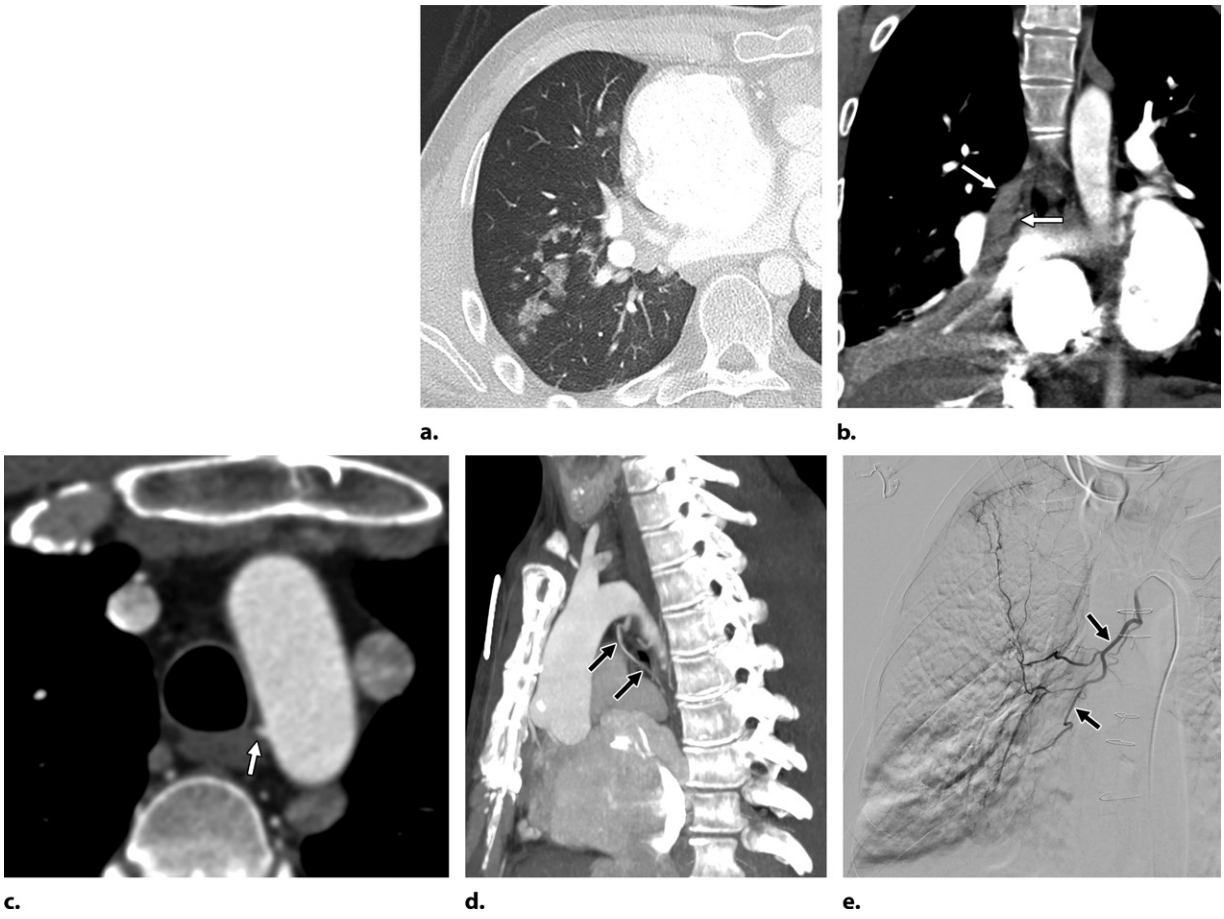


Figure 3. Repaired congenital heart disease manifesting with hemoptysis in a 36-year-old man. **(a)** Axial CT image (lung window) shows ground-glass nodules in the lateral segment of the right middle lobe. **(b)** Coronal CT image (soft-tissue windows) shows a high-attenuation (60 HU) filling defect in the bronchial tree of the right lower lobe (arrows) that is consistent with an intrabronchial clot, which is causing subtotal atelectasis of the right lower lobe. These findings suggest a right-sided source of hemoptysis. **(c)** Axial CT image shows an ectopic bronchial artery arising from the undersurface of the aortic arch, with an enlarged (3 mm) ostium (arrow). **(d)** Sagittal oblique maximum intensity projection multiplanar reconstruction CT image shows that the artery (arrows) is hypertrophied throughout its proximal course. The ectopic bronchial artery was followed to the region of the clot in the bronchus of the right lower lobe. Left-sided bronchial arteries were identified (not shown), but they were normal in caliber. These findings implicated the right bronchial artery as the source of the hemoptysis through the mechanism of chronic systemic vascular hypertrophy. **(e)** Catheter angiogram with injection of the ectopic right bronchial artery (arrows) shows the hypertrophy, and extension of the right bronchial artery to serve the right apical pleura was also observed. No extravasation was seen. The artery was embolized with 500–700-μm polyvinyl alcohol particles to stasis, and the patient's hemoptysis resolved.

Mechanisms of Hemoptysis

No single pathophysiologic mechanism causes hemoptysis. Broadly, the mechanisms can be divided between primary vascular and secondary vascular causes. Primary vascular mechanisms include chronic systemic vascular hypertrophy, focally damaged vessels, dysplastic lung parenchyma with systemic arterial supply, arteriovenous malformations and fistulas, and capillary-level bleeding. Mechanisms in which vessels are secondarily involved include those that directly destroy the parenchyma such as infections and infarcts and those that involve direct invasion of the airway by a tumor or benign process such as a broncholith. Some underlying disease processes can produce hemoptysis by means of multiple mechanisms. For example, *Mycobacterium tuberculosis* infection can cause hemoptysis by means of focal parenchymal destruction and necrosis through chronic systemic vascular hypertrophy in patients with resultant fibrosis and bronchiectasis, through direct vascular damage in patients with a pulmonary artery pseudoaneurysm, and through erosion of a calcified granulomatous lymph node into the airway (7,12). These mechanisms require different treatments.

Focal parenchymal destruction is typically treated with antituberculous medicine and supportive care. Chronic systemic vascular hypertrophy may be amenable to regional endovascular therapy, most often bronchial artery embolization. Directly damaged vessels that develop pseudoaneurysms may require local endovascular therapy or surgery. Finally, airway erosion from a calcified granulomatous lymph node requires either bronchoscopic or surgical treatment.

RadioGraphics

Primary Vascular Mechanisms of Hemoptysis

Chronic Systemic Vascular Hypertrophy.—The lungs have a dual vascular supply that involves deoxygenated blood with lower pressure from the pulmonary arteries and oxygenated blood with higher systemic pressure from the bronchial arteries. The pulmonary arteries are the dominant source of blood to the lungs, supplying 99% of blood flow and participating in gas exchange. The bronchial arteries do not normally participate in gas exchange; their purpose is to supply nourishment to the lung parenchyma and other structures in the mediastinum (1,4). Bronchial arteries course distally and form ramifications along the airway and with the vasovasorum of the pulmonary arteries. In addition, bronchial arteries have normal anastomoses with the pulmonary arteries (13,14).

In conditions that result in chronic hypoxia or inflammation and in neoplasms, cytokines such as angiopoietin-1 and epidermal growth factor can stimulate bronchial and nonbronchial systemic artery dilatation, remodeling, and angiogenesis, a process that is often collectively referred to as *vascular hypertrophy* or *hypertrophy of vascular supply* (4,5,15,16). These hypertrophied arteries can develop more extensive systemic pulmonary arterial anastomoses that participate in gas exchange. When bronchial arteries are hypertrophied, cardiac output to them can increase from a normal output of 1% to up to 30%. Furthermore, these hypertrophied arteries are thin walled and fragile and are exposed to systemic pressure, making them prone to rupture into the airways, which can cause hemoptysis (4,5,16).

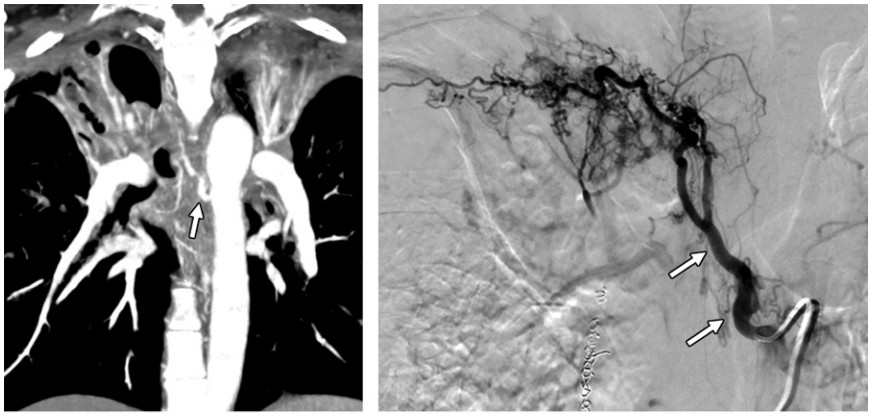
Recognizing chronic systemic vascular hypertrophy is a fundamental goal of CT in patients with hemoptysis. This begins with identification of the bronchial arteries. Bronchial arteries most commonly arise from the proximal descending aorta and are termed *orthotopic* when they originate between the superior endplate of T5 and the inferior endplate of T6. *Ectopic bronchial arteries*, which are common, originate from other systemic arteries, including elsewhere along the aorta or from the subclavian artery, brachiocephalic trunk, thyrocervical trunk, internal mammary artery, or in rare cases, from a coronary artery (16–20). Bronchial arteries, particularly the right bronchial arteries, can also share an origin with an intercostal artery. This configuration, known as an intercostal-bronchial artery trunk, is seen in a majority of patients (17,21). Bronchial artery branching patterns are highly variable. In a cadaver study, Cauldwell and Siekert (22) demonstrated that the most common branching pattern (41% of patients in this study) involved one right bronchial artery arising from an intercostal-bronchial artery trunk and two left bronchial arteries arising from the aorta (22).

Normal bronchial arteries can be identified at CT and are shown to advantage during the systemic arterial phase. Bronchial arteries are typically small serpentine vessels measuring less than 2 mm proximally and less than 0.5 mm distally as they course toward the hila and enter the bronchopulmonary segment, traveling along the bronchial tree. Orthotopic bronchial arteries typically travel posterior to the tracheobronchial tree in the mediastinum. Despite arising elsewhere, ectopic bronchial arteries also ultimately enter the hila and follow a similar distal course along the bronchial tree. Bronchial arteries are considered hypertrophied when they measure larger than 2 mm proximally. The upper limits of normal diameter for middle to distal bronchial arteries are less well defined, but 0.5 mm is a conservative estimated diameter at the level of the hila and distally (17,18).

Depending on the underlying cause, bronchial artery hypertrophy can be unilateral or bilateral. Thus, asymmetric enlargement of bronchial arteries at CT should be regarded as suspicious for hypertrophy (Figs 3, 4). In rare cases, hypertrophied bronchial arteries have an aberrant course that traverses a bronchus; recognizing this finding is important to prevent injury during any attempted bronchoscopy (Fig 5). In patients with hemoptysis, hypertrophied bronchial arteries should be identified, including their origins and course, with careful attention to the presence of ectopic bronchial arteries, because they may require modification of any endovascular treatment.

Multiplanar, maximum intensity projection, and volumetric reconstructions can be of benefit in showing the anatomic course of the hypertrophied bronchial arteries and may serve as a road map for endovascular intervention. In cases of hemoptysis due to hypertrophied bronchial artery supply, contrast material extravasation is not an expected finding.

Although bronchial arteries may contribute to increased systemic vascular supply via hypertrophied normal anatomic pathways, nonbronchial systemic arteries can contribute increased systemic vascular supply via acquired anatomic pathways. Nonbronchial systemic arteries can directly perfuse the lungs via branches entering in pulmonary ligaments and pleural adhesions. These arteries can arise from virtually any systemic vessel in proximity to the thorax and have been reported to arise from the internal mammary arteries, phrenic arteries, intercostal arteries, thyrocervical trunk, costocervical trunk, vertebral arteries, carotid arteries, gastric arteries,



a. b.

Figure 4. Hemoptysis in a 28-year-old man with cystic fibrosis. **(a)** Coronal maximum intensity projection CT image shows upper lobe consolidation and bronchiectasis with volume loss and hilar retraction. Multiple predominantly right-sided enlarged orthotopic bronchial arteries (arrow) are seen in the mediastinum. **(b)** Catheter angiogram with right bronchial artery (arrows) injection shows the same findings as those on the CT image and also shows multifocal contrast material staining in the right upper lobe. These arteries were embolized with 500–700-μm polyvinyl alcohol particles, and the patient's hemoptysis improved.

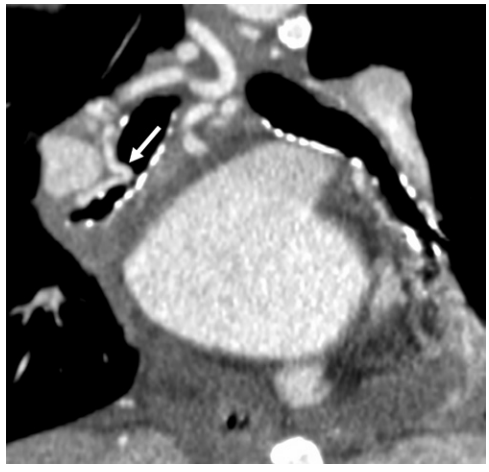


Figure 5. Intrabronchial course of a hypertrophied right bronchial artery in a 90-year-old woman who had a history of chronic mycobacterial infection and presented with hemoptysis. Coronal multiplanar reformation CT image shows numerous enlarged bronchial collateral arteries, one of which courses in the right mainstem bronchus (arrow), implicating this artery as a culprit vessel. This is a direct sign of hemoptysis and is an important finding because recognition of it allows caution during an attempted bronchoscopy to avoid injury.

and coronary arteries. Although differentiation of nonbronchial systemic arteries and ectopic bronchial arteries may be difficult and ultimately not essential, nonbronchial systemic arteries typically do not follow bronchial anatomy and may directly enter the lung parenchyma (4,16,18,23).

Unlike for bronchial arteries, there is no size threshold for an abnormal nonbronchial systemic artery. At CT, simply identifying these arteries or their tributaries entering the lung or extending along the bronchial tree implies recruitment of nonanatomic systemic collateral vessels. Asymmetric enlargement of paired arteries such as intercostal arteries may also be of value in identifying hypertrophied nonbronchial systemic arterial supply. In patients with hypertrophied intercostal arteries, associated pleural thickening may be seen (Fig 6). Moreover, some lesions may have bronchial and nonbronchial systemic arterial supply (ie, dual supply). Thus, even when hypertrophied bronchial arteries are found, a detailed interrogation of potential additional nonbronchial systemic arterial supply should be performed at CT. This is particularly relevant in patients who present with recurrent hemoptysis after initial bronchial artery embolization and may have previously undiagnosed or newly acquired nonbronchial systemic arterial supply (24) (Fig 7).

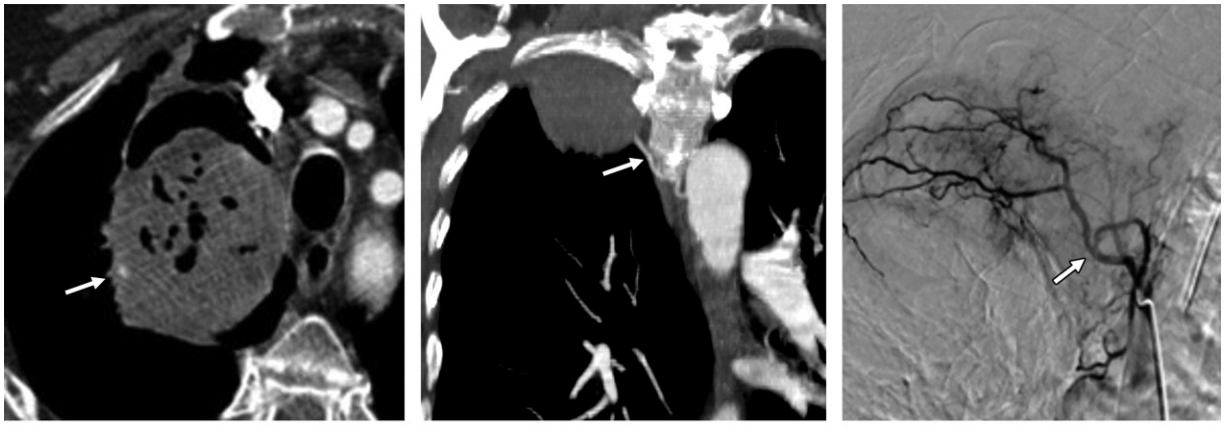
In patients with hemoptysis, identifying nonbronchial systemic arterial supply at CT and reporting their vessels of origin, course, and number are important because these findings can change the endovascular approach. Complex nonbronchial systemic collateral networks may require proximal endovascular therapy or surgery. In some cases, preoperative embolization may also be helpful in controlling anticipated blood loss at surgery (Fig 8). Figure 9 provides a summary diagram of common bronchial and nonbronchial systemic arterial sources of hemoptysis.

Systemic vascular hypertrophy (bronchial and nonbronchial) can be implicated as a cause of hemoptysis in numerous conditions. These include



a. b.

Figure 6. Nonbronchial systemic arterial supply in a 35-year-old man with a history of chronic infection in the left upper lobe who presented with hemoptysis. **(a)** Coronal multiplanar reconstruction CT images show pleural thickening (arrowheads, left image), which is an indirect sign of hemoptysis. Extensive dilated and tortuous arteries (arrow in both images) are seen coursing along the pleural thickening and are consistent with hypertrophied nonbronchial systemic arteries and probable sources of hemoptysis in the left lung. **(b)** Catheter angiograms of the left internal mammary (right image) and lateral thoracic arteries (left image) show collateralization to the left upper lobe via tortuous transpleural collateral vessels, with increased parenchymal staining (arrows). These findings allow confirmation of systemic arteries as the source of the hemoptysis. This patient underwent embolization with a gelatin sponge as a temporizing measure before lobectomy was performed.



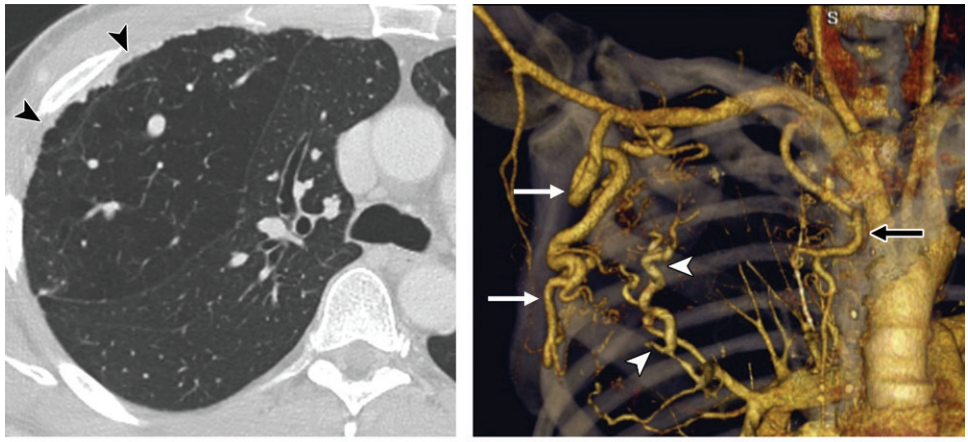
a. b. c.



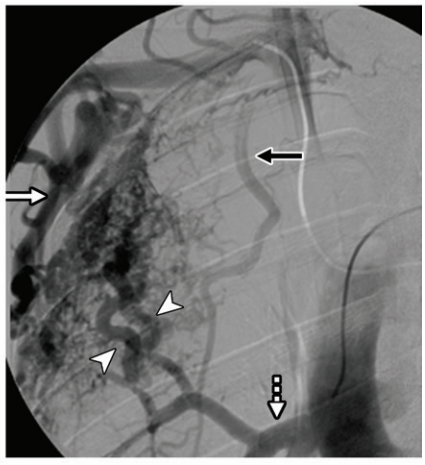
d.

Figure 7. Recurrent hemoptysis in a 74-year-old woman with dual supply to a mycetoma. **(a)** Axial CT image shows a cavity in the right upper lobe that contains internal debris, which is consistent with a mycetoma. Peripheral foci of high attenuation (arrow) are consistent with hypertrophied feeding arteries, but it is difficult to trace them to proximal vessels. **(b)** Coronal multiplanar reformation maximum intensity projection CT image shows a hypertrophied right bronchial artery arising from an intercostal-bronchial artery trunk (arrow). **(c)** Initial catheter angiogram shows that this lesion is primarily served by the right-sided intercostal-bronchial artery trunk (arrow), with collateralization to the lesion via the bronchial trunk. The intercostal-bronchial artery trunk also serves two intercostal arteries. Embolization of the bronchial artery trunk was performed with 300–500-μm polyvinyl alcohol particles. Flow was no longer seen to the lesion after embolization, but the patient had recurrent hemoptysis. **(d)** Original axial maximum intensity projection CT image shows an asymmetrically enlarged right internal mammary artery (black arrow) and costocervical trunk (white arrow), the latter of which terminates in the lesion and both of which had small tributaries in the periphery of the mycetoma (not shown). Repeat angiography (not shown) with selection of the internal mammary artery and costocervical trunk allowed confirmation of the nonbronchial systemic arterial supply to the mycetoma in the right upper lobe. The internal mammary artery was embolized with 500-μm polyvinyl alcohol particles, but the costocervical trunk could not be embolized because vessels serving the cervical spinal cord were observed. With supportive care, the patient's hemoptysis improved after repeat embolization.

Figure 8. Complex vascular supply in a 34-year-old man who had a prior inhalational injury and presented with recurrent hemoptysis. **(a)** Axial CT image (lung window) shows the hyperlucent lung parenchyma in the right upper lobe, with surrounding pleural thickening (arrowheads), containing internal enhancing tortuous vessels that are suggestive of nonbronchial hypertrophied systemic arterial supply. **(b)** Volumetric reconstruction CT image shows to advantage the complex vascular supply, with nonbronchial systemic arterial contributions from the lateral thoracic (white arrows) and right internal mammary (black arrow) arteries. In addition, tortuous enlarged arteries arising from the bronchial arterial tree (arrowheads) are seen extending into the right upper lobe. **(c)** Catheter angiogram of a right subclavian artery injection shows extensive hypertrophied supply from the lateral thoracic artery (solid white arrow) and the internal mammary artery (black arrow). Filling of the right pulmonary arterial tree (dashed arrow) due to backfilling across the enlarged bronchial arteries (arrowheads) is also seen. The lateral thoracic artery contribution to the right upper lobe was embolized prior to surgical resection to prevent excessive bleeding.



a. **b.**



c.

chronic diseases that result in hypoxia, inflammation, or bronchiectasis such as cystic fibrosis, pulmonary fibrosis, congenital heart disease, proximal interruption of the pulmonary artery, chronic pulmonary thromboembolic disease, large-vessel vasculitis, fibrosing mediastinitis, and infection such as with *M tuberculosis*, nontuberculous mycobacteria, and saprophytic *Aspergillus* or mycetoma (17). Tumors can also result in neovascularity and hypertrophied systemic arterial supply, which may be due to angiogenesis, concomitant infection, and/or proximal pulmonary arterial obstruction (17,25,26). The individual CT findings of these conditions are variable, described elsewhere, and beyond the scope of this article. When hemoptysis manifests in patients with these diagnoses or with imaging findings supporting these diagnoses, careful interrogation for and characterization of hypertrophied systemic vascular supply at CT are necessary.

Vascular Damage or Pseudoaneurysm.—A pseudoaneurysm resulting from vascular damage can cause hemoptysis. Pseudoaneurysms are not contained by all the layers of the normal vascular wall, are unstable, and can bleed or rupture into the lung parenchyma or bronchial tree. Pseudoaneurysms are more likely to originate in the pulmonary arterial tree, although bronchial artery and nonbronchial systemic pseudoaneurysms can occur (27–32). At CT, pseudoaneurysms can manifest as a focally dilated vessel or saccular outpouching from a vessel. Pseudoaneurysms may have an irregular wall. When they arise from a small or peripheral vessel, the source vessel may not be definable at CT, and the pseudoaneurysm will only appear as a rounded or irregular focus of contrast material. Direct abutment of a pseudoaneurysm with an airway can be regarded as a finding that is highly suggestive of a vascular to bronchial fistula and a source of hemoptysis. Patients with pseudoaneurysms often bleed intermittently; thus, extravasated contrast material is not required to implicate a pseudoaneurysm as a source of hemoptysis. On rare occasions, extravasated contrast material can be seen arising from a pseudoaneurysm in patients with hemoptysis.

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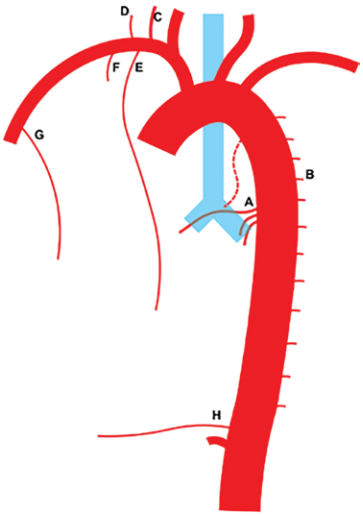
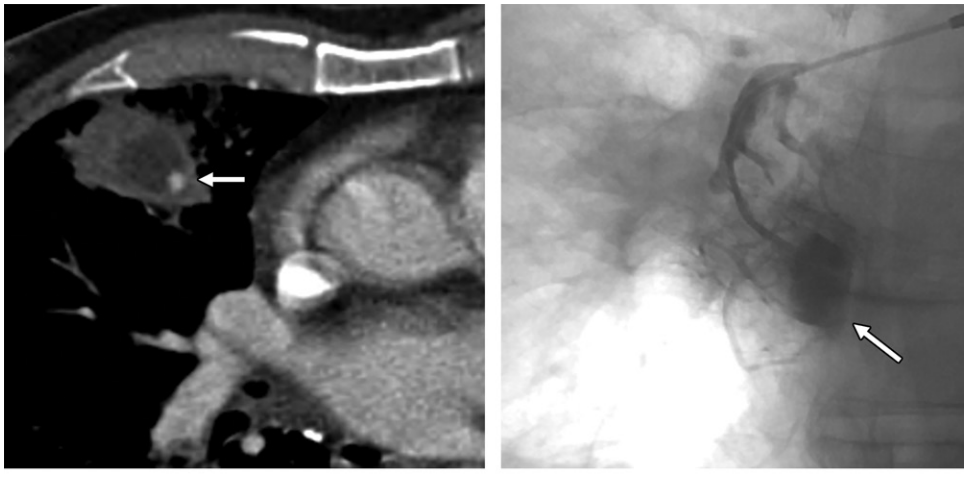


Figure 9. Diagram shows common bronchial and nonbronchial systemic arterial sources of hemoptysis, which are highly variable among patients. *A,* The most common branching pattern for orthotopic bronchial arteries, with one right and two left bronchial arteries arising at the T5-T6 level, is shown. An ectopic bronchial artery (dashed red line) arises from the undersurface of the aortic arch. These variants may also arise from other systemic arteries. *B,* Posterior intercostal arteries arise at each level and course through the intercostal spaces to anastomose with anterior intercostal arteries (not shown). Several branches of the proximal or midsubclavian artery may give rise to transpleural collateral vessels including, *C,* the vertebral artery; *D,* the thyrocervical trunk; *E,* the internal mammary artery; and, *F,* the costocervical trunk. *G,* The lateral thoracic artery traverses the lateral chest wall and may also produce transpleural collateral vessels. *H,* Inferior phrenic arteries may arise directly from the abdominal aorta or as branches from the celiac axis and can occasionally be a source of hemoptysis, primarily in collateral vessels through the inferior pulmonary ligament.



a. **b.**

Figure 10. Pulmonary artery infectious pseudoaneurysm arising from direct extension of angioinvasive pneumonia in a 42-year-old man with neutropenia who presented with hemoptysis and had multiple necrotizing nodules that were consistent with angioinvasive *Aspergillus*. **(a)** Axial CT image shows a small focus of contrast material (arrow) with surrounding low attenuation along the periphery of a nodule in the right middle lobe, which are findings consistent with a partially thrombosed infectious pseudoaneurysm arising from the pulmonary arterial tree. **(b)** Subsequent catheter angiogram with right middle lobe artery injection shows the pulmonary artery origin and the presence of the pseudoaneurysm (arrow). The lesion was embolized with microcoils.

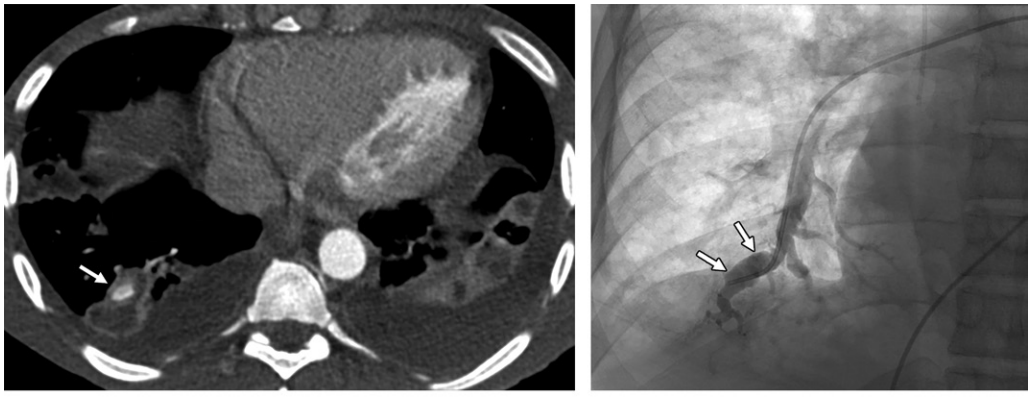
Pulmonary artery pseudoaneurysms have several underlying causes, but infection is the most common (28). Direct extension from adjacent or surrounding pneumonia is one means of developing an infectious pseudoaneurysm, in which the vascular wall is seeded from the outside in. This can occur with any type of pneumonia, although it is more often seen in patients with necrosis or an abscess. A classic example of an infectious pseudoaneurysm arising next to pneumonia is a Rasmussen aneurysm, a rare complication of *M tuberculosis* infection.

At CT, a Rasmussen aneurysm manifests as a rounded focus of enhancement of variable size along the periphery of an area of necrosis due to *M tuberculosis*. Because this is an acute process, a Rasmussen aneurysm arises from a typically small but otherwise normal vessel (28,33–35). Thus, in many cases, it may not be traceable to a vessel of origin. Similar pseudoaneurysms can be seen on the periphery of other infections, including those of fungal or bacterial origin (28) (Fig 10). Infectious pseudoaneurysms can also arise in the central aspect of pneumonia, again more commonly if there is associated necrosis or an abscess (Fig 11).

Infectious pulmonary artery pseudoaneurysms can also develop in patients with septic pulmonary embolic disease. Any patient with parenchymal findings of septic emboli, including rounded, peripheral, and often cavitary pulmonary nodules and wedge-shaped opacities, can develop an



Figure 11. Infectious pseudoaneurysm in a 62-year-old man with necrotic right lower lobe pneumonia who developed hemoptysis. Axial CT image shows an ovoid focus of high attenuation (arrow) in the necrotic pneumonia that enhances similar to the pulmonary arterial tree. This finding was consistent with an infectious pulmonary artery pseudoaneurysm. The patient was treated with coil embolization.



a. **b.**

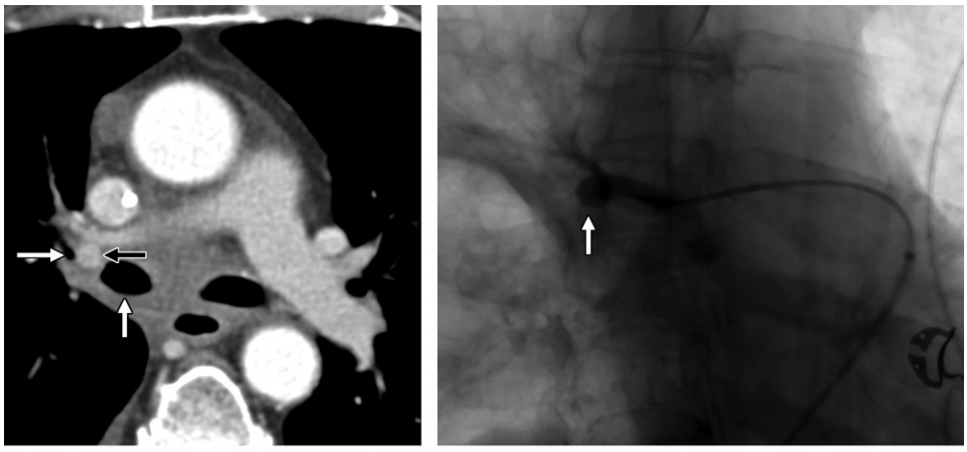
Figure 12. Pulmonary artery pseudoaneurysm in a 42-year-old man with septic embolic disease who developed hemoptysis. **(a)** Axial CT image shows multiple peripheral abscesses that are consistent with the patient's known diagnosis. In the right lower lobe, a rounded focus of contrast material (arrow), which is consistent with an infectious pseudoaneurysm, is seen. **(b)** Catheter angiogram shows the pseudoaneurysm (arrows), which was treated with coil embolization. In patients with septic embolic disease, a careful search for infectious pseudoaneurysms is required because their presence may change patient treatment by requiring embolization or, in some cases, resection.

associated infectious pulmonary artery pseudoaneurysm. In a patient with septic emboli, infectious pseudoaneurysms are also particularly prone to occur in vessels where macroscopic infectious emboli have lodged. Thus, focal dilatation of a pulmonary artery that contains an infectious embolism should be considered an infectious pseudoaneurysm. Patients with septic embolic disease can have solitary or multiple infectious pseudoaneurysms that often enlarge rapidly (28) (Fig 12).

Noninfectious pulmonary artery pseudoaneurysms have several potential causes, including direct invasion from tumors; inflammation related to vasculitis, radiation therapy, or blunt or penetrating trauma; and iatrogenic causes including the use of Swan-Ganz catheters and the performance of percutaneous biopsies (28,33). Pulmonary artery pseudoaneurysms, whether infectious or noninfectious, cause the minority of cases of hemoptysis but can result in severe life-threatening hemoptysis when they rupture.

Despite these potentially catastrophic outcomes and the reality that they are readily detectable at CT, pulmonary artery pseudoaneurysms are frequently missed (6,28,36). This is typically not due to technical factors, because the pulmonary arteries should be opacified at CT during the systemic arterial phase without a saline chaser, but rather, it is due to a lack of close interrogation of the pulmonary arteries. Thus, particularly in cases of hemoptysis, diagnostic radiologists should look carefully for pulmonary artery pseudoaneurysms because their presence changes patient treatment options. Pulmonary artery pseudoaneurysms are most commonly treated with coil embolization and require venous rather than systemic arterial access. To aid in treatment planning, when pulmonary artery pseudoaneurysms are detected, the location, size, and relationship to the bronchial tree should be reported (Fig 13).

Although uncommon, systemic arterial pseudoaneurysms and true aneurysms can also result



a. b.

Figure 13. Postradiation tumor pseudoaneurysm in a 49-year-old woman with a history of radiation therapy–treated lung cancer who presented with hemoptysis. Initial interpretation of the patient's CT examination showed normal bronchial arteries and no systemic arterial explanation for hemoptysis. The patient underwent prophylactic bronchial artery embolization. **(a)** Repeat axial CT image shows persistent hemoptysis and a rounded pulmonary artery pseudoaneurysm (black arrow) in the right upper lobe in the radiation-treated tumor that directly abuts the proximal and distal branches of the bronchial tree (white arrows), implicating it as the source of hemoptysis. **(b)** Pulmonary artery angiogram shows the pseudoaneurysm (arrow). Microcoils were used for definitive treatment. This case underscores the necessity of careful interrogation of the pulmonary arteries in patients with hemoptysis, particularly in those with a history of a neoplasm or with normal bronchial arteries. Pulmonary artery pseudoaneurysms are frequently missed and can be a source of hemoptysis in patients who have already undergone bronchial artery embolization.

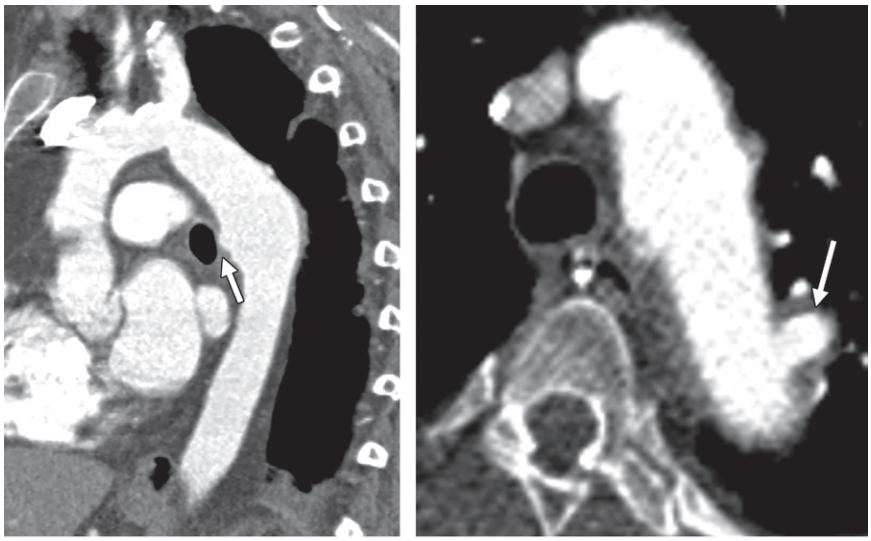


a. b.

Figure 14. Unstable aortic aneurysm causing hemoptysis in a 51-year-old man who presented with chest pain. **(a)** Coronal CT image shows a large unstable saccular aneurysm (arrow) of the descending aorta. **(b)** Axial CT image (lung window) shows surrounding ground-glass opacity (arrow), which is consistent with associated pulmonary hemorrhage and implicates the aneurysm as the source of the hemoptysis. This patient was treated with endovascular stent repair.

in hemoptysis. These are most commonly aortic in origin and may be due to rupture, postsurgical complications, or infection (Fig 14). In rare cases, bronchial artery pseudoaneurysms can occur (37). CT findings implicating a systemic arterial pseudoaneurysm as a cause of hemoptysis include direct abutment with a bronchus or extension directly into the lung parenchyma, which are findings suggestive of a systemic artery to bronchial fistula (Fig 15). Patients with systemic artery to bronchial fistulas bleed intermittently, and thus, contrast material extravasation across a systemic artery to a bronchial fistula is an uncommon CT finding (Fig 16). Systemic arterial pseudoaneurysms resulting in hemoptysis may be treated with endovascular techniques or surgery, depending on their size, shape, and location (29–32).

Dysplastic Lung Parenchyma with Systemic Arterial Supply.—*Pulmonary sequestrations* are defined by dysplastic nonfunctioning pulmonary tissue that is not normally connected to the bronchial tree. Pulmonary sequestrations have systemic



a. b.

Figure 15. Aortobronchial fistulas in two patients. **(a)** Sagittal oblique CT image in a 51-year-old man who presented with hemoptysis many years after coarctation repair shows localized enlargement of the descending aorta, with a focal outpouching that is consistent with a pseudoaneurysm (arrow), and an aortobronchial fistula that abuts the left mainstem bronchus. These findings are suggestive of an aortobronchial fistula, which was confirmed at surgical repair. **(b)** Axial CT image in a 59-year-old woman who presented many years after aortic arch repair with intermittent hemoptysis shows a focal outpouching from the proximal descending aorta (arrow), which is protruding into the left upper lobe and is enlarged in comparison to its appearance at prior examinations. Associated ground-glass opacity in the left upper lobe (not shown) was suggestive of hemorrhage. These findings are suggestive of an aortic to bronchial fistula, and the diagnosis was confirmed at surgical repair.

arterial supply that is most often from the thoracic or abdominal aorta. Pulmonary sequestrations can also derive their systemic arterial supply from other systemic arteries, including the celiac artery, splenic artery, intercostal arteries, subclavian arteries, and coronary arteries. In some cases, the systemic arterial supply is complex and from multiple vessels. There are two main types of sequestrations. Extralobar sequestrations, which account for approximately 25% of cases, are thought to be congenital, are encased by their own pleura, and have systemic drainage. Intralobar sequestrations, which typically drain to the pulmonary veins, have a more controversial origin, and in many cases, may be acquired after repeated bouts of pneumonia that led to parasitization of the adjacent systemic vasculature (38–40).

Most patients with pulmonary sequestrations present with recurrent infections, but hemoptysis can also occur (38,41–43). The mechanism for hemoptysis is thought to be related to the higher systemic pressures of the feeding arteries, which may also be fragile due to medial hypertrophy and endarteritis (38,42). At CT, pulmonary sequestrations have variable features including solid enhancing portions, cystic or emphysematous change, and cavitation. The associated bronchi may be dilated or filled with mucus. The key diagnostic feature of pulmonary sequestration at CT is identification

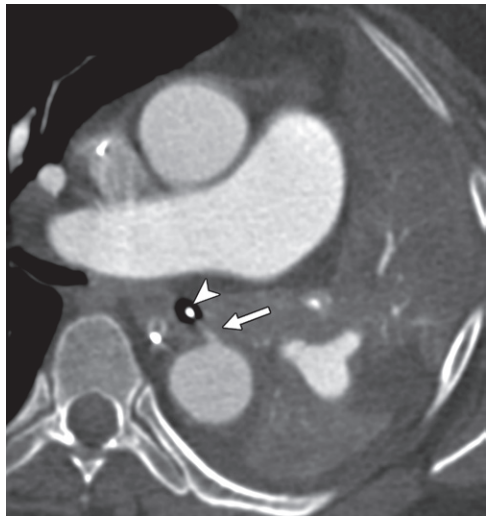


Figure 16. Life-threatening hemoptysis in a 68-year-old woman with a history of acute myelogenous leukemia. The hemoptysis was treated bronchoscopically with placement of a balloon occluder, which caused collapse of the left lung. Axial CT image from a subsequent examination shows extravasation of contrast material from the aorta toward the left mainstem bronchus (arrow), which is a rarely seen finding in patients with an aortobronchial fistula. The intrabronchial catheter is also seen (arrowhead). This patient was treated with placement of an endovascular stent of the aorta.

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of the systemic feeding arteries (38–40). Thus, when an abnormal area of the lung parenchyma, particularly in the lower lobes, is present, careful interrogation should be performed for anomalous systemic vascular supply. When patients with sequestrations present with hemoptysis, findings of internal parenchymal hemorrhage, including airspace and nodular opacities, can be identified and thus can implicate the sequestration as the source. Treatment of patients with a pulmonary sequestration complicated by hemoptysis is mainly surgical, although preoperative embolization of the systemic arterial supply may be performed to prevent blood loss (41,42) (Fig 1).

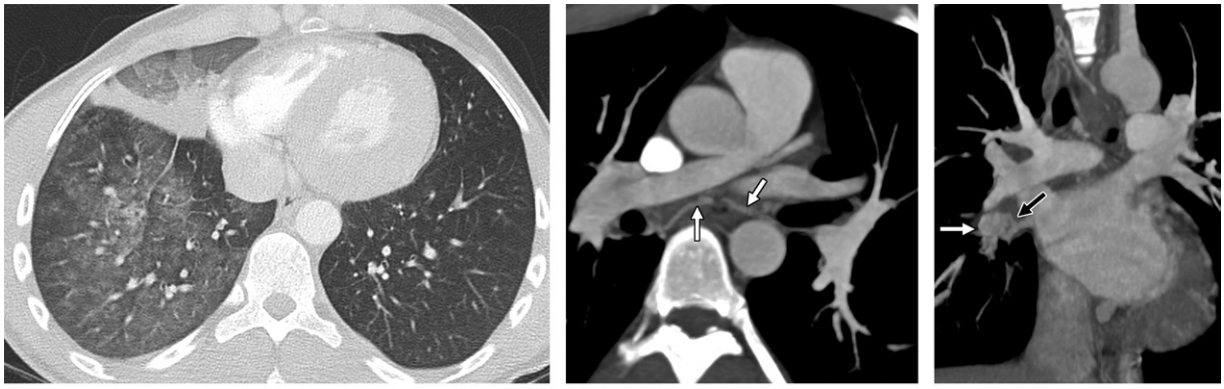
Arteriovenous Malformations and Fistulas.— Arteriovenous malformations are abnormal connections between arteries and veins via a thin-walled aneurysm or nidus. In the thorax, pulmonary arteriovenous malformations are most common and represent abnormal connections between pulmonary arteries and veins. Pulmonary arteriovenous malformations are typically congenital and are usually part of a syndrome such as hereditary hemorrhagic telangiectasia. In studies from referral centers (44–47), more than 90% of patients with pulmonary arteriovenous malformations had hereditary hemorrhagic telangiectasia. Up to 50% of patients with hereditary hemorrhagic telangiectasia have pulmonary arteriovenous malformations. Systemic feeding arteries can also contribute to arteriovenous malformations, with sources including the bronchial, aortic, intercostal, and internal mammary arteries.

Arteriovenous fistulas, which are acquired connections between arteries and veins, are most often traumatic or iatrogenic in origin. A related condition, a bronchial artery–to–pulmonary artery fistula, can also occur and behaves similarly to traditional arteriovenous malformations. We do not regard bronchial artery–to–pulmonary artery fistulas as true malformations or fistulas, but rather as hypertrophy of the normal bronchial artery–to–pulmonary artery anastomoses due to congenital or unknown acquired causes (13,48). The mechanism for bleeding in arteriovenous malformations and fistulas is due to rupture of the thin-walled connections and involved vessels that can enlarge over time, particularly when they have a systemic arterial supply and are thus subject to higher levels of pressure (49–52). Many arteriovenous malformations and more than 99% of pulmonary arterial feeding vessels do not result in hemoptysis and may be discovered incidentally (50).

At CT, arteriovenous malformations can be identified by recognizing the enlarged typically tortuous feeding artery and enlarged draining vein. Arteriovenous malformations can have variable configurations ranging from simple connections to a small aneurysm or nidus to complex vascular networks with multiple feeding arteries. When CT is performed in an early arterial phase, early venous filling may be seen in association with an arteriovenous malformation. In patients who present with hemoptysis, an arteriovenous malformation can be implicated if there are findings of pulmonary hemorrhage in the surrounding lung parenchyma. For more central arteriovenous malformations, direct abutment of the nidus with a bronchus or encasement of the bronchial tree also supports malformation as the cause. In addition, findings of distally aspirated blood may be seen (Fig 17).

When arteriovenous malformations are found at CT, the reader should report the location, source (specific artery and whether pulmonary or systemic), and size of the feeding artery and the distribution of the draining veins. Pulmonary arterial venous malformations, when they are amenable (ie, those with a feeding artery greater than 3 mm in diameter), are typically preemptively treated endovascularly with microcoils to prevent paradoxical embolization. Endovascular therapy is also the first-line treatment of arteriovenous malformations, whether pulmonary or systemic, that cause hemoptysis. For patients who cannot safely undergo embolization or for whom embolization was not definitive, surgical resection may be necessary (49,51,52).

Capillary-level Bleeding.—Capillary-level bleeding can result in diffuse alveolar hemorrhage (DAH), which is characterized by extravasation of red blood cells into the pulmonary alveolar spaces. The extravasated red blood cells can result in hemoptysis, but up to 33% of patients with DAH do not present with clinical hemoptysis (53). DAH can be the result of immune-mediated and nonimmune-mediated causes. Immune-mediated causes, including primary and secondary small-vessel vasculitides, are most common. Nonimmune causes (ie, the use of anticoagulation therapy or mitral stenosis) can also result in DAH (53,54). At CT, DAH manifests with findings of parenchymal blood products that appear as a combination of ground-glass opacity and consolidation. These findings typically suggest a differential diagnosis that includes pneumonia and edema but should be regarded as concerning for pulmonary hemorrhage in a patient with hemoptysis (55). DAH is often diffuse, particularly when it is due to underlying systemic causes, but can be focal or patchy. When DAH is encountered at CT, careful attention should be directed toward identifying clues to its underlying cause. For example, the



a. b. c.

Figure 17. Bronchial artery to pulmonary vein malformation in a 19-year-old man with intermittent hemoptysis for 10 days who received a diagnosis of pneumonia as an outpatient. His hemoptysis worsened, and he presented to the emergency department . His hemoglobin level was 6.7 g/dL. **(a)** Axial CT image (lung window) shows extensive nodular ground-glass opacity in the right lower and middle lobes, with atelectasis in the right middle lobe. In a patient with hemoptysis, these findings are consistent with aspirated right-sided blood products. **(b)** Axial maximum intensity projection CT image (soft-tissue windows) shows a hypertrophied right bronchial artery (arrows). **(c)** Coronal maximum intensity projection CT image shows that this hypertrophied right bronchial artery extends into a tangle of dilated vessels in the right hilum (white arrow), encasing the right bronchial tree. This tangle of vessels connects with the right inferior pulmonary vein in multiple locations, one of which is shown (black arrow). These findings are consistent with a bronchial artery to pulmonary vein malformation. Emergency angiography was performed, but a safe route for embolization could not be found because of the complexity of the lesion and the large connections with the pulmonary veins. The patient was at risk for systemic embolization, became hemodynamically unstable, and ultimately required a right middle and lower lobectomy for definitive treatment.

concomitant presence of cavitary nodules or vascular wall thickening implicates vasculitis as the source of DAH. In many cases, the cause of DAH is determined on the basis of laboratory test results and pathologic results of biopsies. Treatment of DAH is centered on supportive care and addressing the underlying cause (Fig 18).

Secondary Vascular Mechanisms of Hemoptysis

Parenchymal Destruction.—Destruction of the pulmonary parenchyma is an important secondary vascular mechanism for hemoptysis. The most common cause of parenchymal destruction resulting in hemoptysis is acute infectious pneumonia. Although pneumonia due to any infectious agent can result in hemoptysis, tuberculous mycobacterial, nontuberculous mycobacterial, and angioinvasive fungal infections are more commonly associated with hemoptysis. Bacterial pneumonia can also result in hemoptysis, particularly when there is associated necrosis or abscess formation (56–60).

At CT, findings of infectious pneumonia vary and include nonenhancing areas of parenchymal consolidation, nodules, and ground-glass opacity. Findings of necrosis including central low-attenuation phlegmonous material and gas indicate extensive parenchymal destruction. Abscess formation can occur, in which a region of infectious necrosis develops a well-defined wall. Ground-glass opacity surrounding an infectious pulmonary nodule or consolidation (the ground-glass "halo" sign) is suggestive of surrounding pulmonary hemorrhage and has been described specifically in angioinvasive fungal infections.

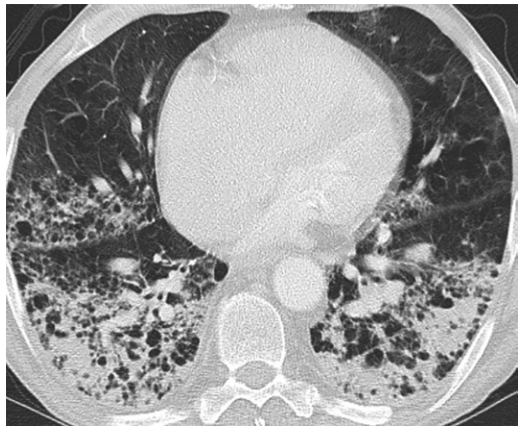


Figure 18. Hemoptysis in a 55-year-old man who was taking anticoagulant medication. Axial CT image shows dependent bilateral lower lobe airspace opacification superimposed on emphysema, which is consistent with diffuse alveolar hemorrhage. The diagnosis was later confirmed at bronchoscopy. No specific causative lesion or abnormal vessel was identified, so the patient's condition was managed conservatively.

It is important to recognize that parenchymal destruction is one mechanism by which an infectious pneumonia can result in hemoptysis. As we previously discussed, infections can also cause hemoptysis acutely by means of pseudoaneurysm formation or chronically through systemic vascular hypertrophy. These different mechanisms have important treatment implications. Pseudoaneurysms and systemic vascular hypertrophy typically require either endovascular

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Figure 19. Hemoptysis in a 61-year-old man with a history of sarcoma and multiple metastases. Axial CT image shows multiple pulmonary nodules, but a larger mass in the right lower lobe was considered the culprit lesion for hemoptysis because the surrounding ground-glass opacity indicates hemorrhage (arrow), which was a new finding when compared with the patient's prior examination findings.

or surgical intervention. Infections causing hemoptysis by means of parenchymal destruction alone are most often treated with antimicrobial agents. In exceptional circumstances in which hemoptysis is unrelenting or an infection that is complicated by abscess cannot be cleared, surgical resection or percutaneous drainage may be pursued (59,60). The diagnostic radiologist must be aware of these different mechanisms, their respective findings at CT, and their implications for treatment.

Tumor necrosis is another important cause of hemoptysis. In a patient with hemoptysis, CT findings of tumor necrosis, including central low attenuation, internal blood products, or cavitation, should cause concern (61,62). CT findings of adjacent or surrounding parenchymal hemorrhage or downstream aspirated blood products can serve as supportive evidence that a tumor is the source of hemoptysis, particularly when a patient has multiple lesions (Fig 19). When tumor necrosis is implicated as a source of hemoptysis, treatment options typically include supportive care or surgery. As with infections, careful interrogation for associated pseudoaneurysms or hypertrophied systemic vascular supply is warranted because these findings may allow endovascular treatment.

Hemoptysis can also be seen in a minority of patients with a pulmonary embolism, but the hemoptysis is typically not life threatening (63–66). Although the mechanism of hemoptysis in patients with a pulmonary embolism may be multifactorial and related to bronchopulmonary shunting, hemoptysis is also seen in patients with pulmonary embolism and accompanying pulmonary infarcts. As such, parenchymal destruction due to a pulmonary infarct may also be a cause of hemoptysis. In either case, diagnostic radiologists should be aware that a pulmonary embolism and pulmonary infarcts can result in hemoptysis. Parenchymal findings of pulmonary infarcts, which include peripheral consolidation, ground-glass opacity, and areas of consolidation with central clearing (ie, the "reverse-halo" sign), should prompt investigation for pulmonary emboli to the involved territory (67). Identification of pulmonary emboli in patients with hemoptysis is important because treatment can then be directed at the pulmonary emboli and may include a combination of medical (ie, anticoagulation therapy) and interventional (eg, filters, clot retrieval, or lysis) techniques.

Airway Invasion.—Direct airway invasion can also result in hemoptysis. Tumors such as bronchogenic carcinoma, carcinoids, and metastases can cause hemoptysis when they invade or arise within the airway (68–70). CT findings of a tumor that arises in, deforms, or directly invades an airway should be regarded as potential sources of hemoptysis (Fig 20). Interrogation of the lung parenchyma distal to the tumor may offer corroborating findings of aspirated blood products. As with pulmonary infections, tumors can cause hemoptysis by means of multiple mechanisms including tumor necrosis, direct vascular damage with pseudoaneurysm formation, and systemic vascular hypertrophy. However, identification of airway invasion at CT is important because this can direct patient treatment to endobronchial and surgical therapy rather than to the vascular-focused therapies that would be used for hemoptysis with other causes.

Broncholiths can also result in hemoptysis by means of direct airway invasion. A calcified lymph node, which can be due to prior infection (eg, tuberculosis or histoplasmosis) or exposure can erode into the bronchial tree. At CT, broncholiths are identified as calcified nodules that are present either entirely within or partly within an airway. When prior examinations are available, broncholiths can be seen to change position over time as they erode into the airway. Increased distal atelectasis can result, and when patients have hemoptysis, there may be findings of aspirated blood products in the distal airways. Identifying broncholiths at CT will direct treatment of hemoptysis to bronchoscopic or surgical therapy (12) (Fig 21).

Cryptogenic Hemoptysis

In patients who present with hemoptysis in whom no purported cause or lesion is identified, a diagnosis of cryptogenic hemoptysis can be made. At CT,

patients with cryptogenic hemoptysis may have lung parenchymal or airway findings of hemoptysis, but no CT or clinical explanation for the findings is found. Cryptogenic hemoptysis represents 10%–20% of patients with hemoptysis and is more commonly seen in patients who smoke (71–73).

Herth et al (73) followed 135 patients with cryptogenic hemoptysis and found that 6% were diagnosed with an unresectable lung cancer within the next 6 years. This work raises the possibility of hemoptysis as an underlying marker of dysplasia or metaplasia that ultimately may be due to a premalignant condition or an occult malignancy.

However, in a different study, Savale et al (74) followed 81 patients with cryptogenic hemoptysis and found no cases of lung cancer. In this study, Dieulafoy lesions, which are vascular anomalies characterized by enlarged bronchial mucosal or submucosal arteries, were found in five of nine patients who subsequently underwent surgery. Dieulafoy lesions are expected to only show findings of aspirated blood at CT, because they are too small to be directly visualized. Thus, although there is not consensus on an association between cryptogenic hemoptysis and malignancy, we recommend exercising caution and performing follow-up CT in 6–12 months in patients diagnosed with cryptogenic hemoptysis to evaluate for an underlying malignancy that may have been occult at previous imaging.

Conclusion

In this article, we have provided an overview of the major mechanisms that can cause hemoptysis and their associated CT findings. These mechanisms and specific diseases that can cause them are summarized in Table 2. To use this information in real clinical practice, a structured yet flexible approach to CT interpretation is required. Our suggested approach is presented in Table 3 and is based on the identification of indirect and direct signs. Indirect pulmonary signs are the CT manifestations of hemorrhage and include both findings of parenchymal bleeding and aspirated blood. Indirect vascular signs are indicative of increased vascular supply. Direct signs are more specific and implicate a focally abnormal causative culprit vessel or lesion. Examples include an infectious pseudoaneurysm (ie, a culprit vessel) or a tumor (ie, a culprit lesion).

The use of indirect and direct signs at CT interpretation has two important functions. First, indirect and direct signs can be used to support an underlying mechanism as the cause of hemoptysis. Some mechanisms, such as systemic vascular hypertrophy, will be described with indirect signs only, because patients present with pulmonary findings indicative of parenchymal bleeding as well as findings of increased vascular supply. Other mechanisms are combinations of indirect and direct signs. For example, a bleeding focally damaged vessel may demonstrate the direct sign of formation of a pseudoaneurysm and an indirect sign of adjacent parenchymal hemorrhage or distally aspirated blood.

Perhaps more relevant, however, is the use of indirect and direct signs identified at CT to

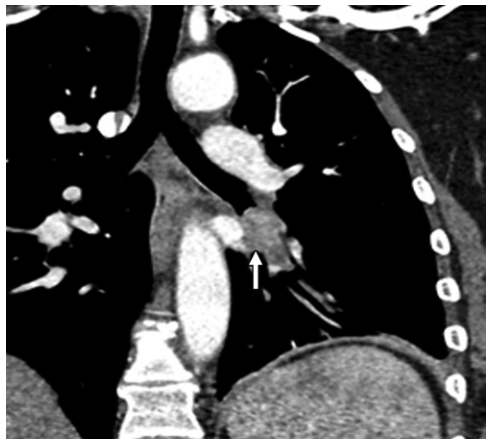


Figure 20. Hemoptysis in a 31-year-old woman. Coronal multiplanar reformation CT image shows an endobronchial mass (arrow) in the left lower lobe bronchus. Given the patient's age, the endobronchial location, and the high attenuation of the lesion, a carcinoid tumor was favored as the source of the hemoptysis. The lesion was surgically resected and proved to be a carcinoid tumor.

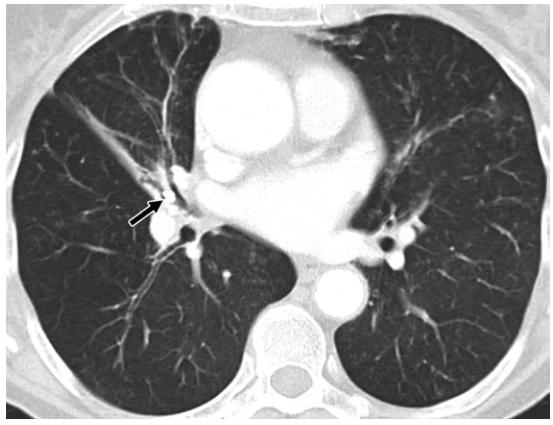


Figure 21. Broncholith causing hemoptysis in a 69-year-old woman with recurrent intermittent hemoptysis. Axial CT image shows a calcified nodule (arrow) in the bronchial tree of the right middle lobe, resulting in partial atelectasis of the lateral segment. These findings are consistent with a broncholith and show that the source of the hemoptysis is the right middle lobe. The right-sided bronchial tree was interrogated with rigid bronchoscopy, which allowed confirmation of the presence of a calcified lesion protruding into and occluding the bronchus. Attempted manipulation of this lesion resulted in additional hemorrhage; thus, a right middle lobectomy was performed. The broncholith was confirmed at pathologic analysis of the resected specimen.

Table 2: Mechanisms and Causes of Hemoptysis

Primary vascular mechanism
Systemic vascular hypertrophy
Malignancy
Cystic fibrosis
Bronchiectasis
Tuberculosis
Nontuberculous mycobacteria
Lung damage from prior infection
Mycetoma
Congenital heart disease
Interstitial lung disease
Fibrosing mediastinitis
Chronic thromboembolic pulmonary disease
Congestive heart failure
Proximal interruption of the pulmonary artery
Focally damaged vessels or pseudoaneurysm
Malignancy
Tuberculosis
Fungal pneumonia
Bacterial pneumonia
Septic pulmonary embolism or infarction
Large- or medium-vessel vasculitis
Blunt and penetrating chest injury
Swan-Ganz catheter perforation
Transbronchial biopsy
Percutaneous lung needle biopsy
Ruptured thoracic aortic aneurysm
Aortobronchial fistula
Airway stent erosion
Radiation
Dysplastic parenchyma with systemic supply caused by pulmonary sequestration
Arteriovenous malformation or fistula
Congenital (eg, hereditary hemorrhagic telangiectasia)
Blunt and penetrating chest injury
Transbronchial biopsy
Percutaneous lung needle biopsy
Thoracentesis or chest tube placement
Capillary-level bleeding
Small-vessel vasculitis
Medications (eg, anticoagulants, antiplatelets)
Illicit drugs (eg, crack cocaine)
Thrombocytopenia
Mitral stenosis
Secondary vascular mechanism
Parenchymal destruction
Malignancy
Tuberculosis
Nontuberculous mycobacteria
Bacterial pneumonia
Fungal pneumonia
Pulmonary embolism or infarction
Airway invasion
Malignancy
Broncholith

Table 3: CT Approach—Direct and Indirect CT Signs in Patients with Hemoptysis

Indirect CT signs
Pulmonary signs*
Centrilobular nodules (including tree-in-bud)
Ground-glass opacities
Consolidation
Septal line thickening when subacute
Vascular signs†
Increase in number of feeding arteries
Increase in size of feeding arteries
Pleural thickening (potentially with traversing vessels)
Direct CT signs
Culprit lesion‡
Infections
Tumors
Broncholiths
Culprit vessel§
Infectious pseudoaneurysm
Tumor pseudoaneurysm
Unstable aortic aneurysm
Aortobronchial fistula

*Findings of parenchymal hemorrhage or aspirated blood.
†Findings of increased vascular supply.
‡Potentially treatable localized source.
§Potentially treatable localized vessel.

help direct patient treatment. Indirect pulmonary signs can be used to localize treatment to a particular portion of the lung, or conversely, when diffuse hemorrhage is present, to indicate that no localizable site of hemorrhage is identifiable. Indirect vascular signs are valuable when endovascular therapies are pursued because they can implicate a specific vascular source. This is particularly important when the source lies outside orthotopic bronchial arteries because the interventional radiologist is directed to vascular sources that may not otherwise have been interrogated. Direct signs identified at CT affect patient treatment by defining a focal targetable cause of hemoptysis. For example, identification of a broncholith (ie, a culprit lesion) drives treatment away from endovascular therapies and toward surgical or bronchoscopic treatment. Recognition of a pulmonary artery pseudoaneurysm, (ie, a culprit vessel) necessitates either surgical treatment or endovascular treatment with a systemic venous approach.

In conclusion, CT has emerged as an important noninvasive tool in the evaluation of patients with hemoptysis. To best interpret these examinations, diagnostic radiologists must understand

the potential mechanisms of hemoptysis and their imaging features. This will allow accurate diagnosis and the identification of CT findings that can be used to direct treatment.

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